

Atul Kumar

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Current Role

Ph.D. Scholar — Data Science & Engineering

2023 – Present

Indian Institute of Science Education and Research (IISER) Bhopal | CGPA: 8.83/10
Thesis: *Privacy in Machine Learning* | Supervisor: Dr. Akshay Agarwal

Education

Integrated M.Sc. in Computer Science

2018 – 2023

Central University of Rajasthan

Intermediate (Class XII)

2017

Bihar School Examination Board

Matriculation (Class X)

2014

Central Board of Secondary Education

Experience

- Teaching Assistant — Deep Learning, IISER Bhopal** Jan 2026 – April 2026
 - Mentored undergraduate and graduate students in neural networks, CNNs, transformers, and modern deep learning architectures.
 - Conducted tutorials, doubt-clearing sessions, and project guidance for course assignments.
- Teaching Assistant — Machine Learning with Graphs, IISER Bhopal** Aug 2025 – Dec 2025
 - Assisted students in understanding graph neural networks, graph representation learning, and network analysis.
 - Supported coursework, programming assignments, and evaluation activities.
- Teaching Assistant — Applied Optimization, IISER Bhopal** Jan 2025 – Apr 2025
 - Guided students in optimization techniques including convex optimization, gradient-based methods, and constrained optimization.
 - Assisted in preparing tutorials and assessing assignments.
- Teaching Assistant — Database Management Systems, IISER Bhopal** Aug 2024 – Dec 2024
 - Mentored students in relational database design, SQL, normalization, and transaction management.
 - Facilitated laboratory sessions and evaluated coursework.

Research Interests

Privacy-Preserving Machine Learning, Trustworthy AI, Computer Vision, Biometric Security, Data & Machine Unlearning, and Responsible AI Systems.

Publications

- A. Kumar and A. Agarwal, *Visible Yet Unrecognizable: Frequency-Selective Facial Privacy via Attention*, European Conference on Computer Vision (**ECCV 2026**), h5-index: 262.
- S. Gehlot, A. Kumar, A. Agarwal, *Beyond Euclidean Space: Hyperbolic Identity Geometry for Aerial-Ground Person Re-Identification*, Women in Computer Vision Workshop @ **ECCV 2026 (WICV)**.
- Mohit, A. Kumar, A. Agarwal, *Faces in the Wild: GAN-Driven Normalization for Robust Facial Recognition*, CVPR Workshop on AI for Creative Visual Content Generation, Editing and Understanding (**CVPRW 2026**).
- R. Jain, A. Kumar, A. Agarwal, *Semantic-Guided Sketch-to-RGB Image Generation via Controlled Diffusion for Improved Sketch Recognition*, AAAI Conf. on Artificial Intelligence (**AAAI 2026**), h5-index: 232.
- R. Jain, A. Agarwal, A. Kumar, *Deep Models Under Domain Shift: A Sketch-Based Study*, Women in Machine Learning Workshop @ **NeurIPS 2025**.

6. **A. Kumar**, A. Agarwal, N. Ratha, *Your Face, Your Privacy: Combating Unauthorized Usage*, IEEE Int. Conf. on Automatic Face and Gesture Recognition (**FG 2025**), h5-index: 40.
7. M. RA*, **A. Kumar***, A. Agarwal, *On Which Data Distribution (Synthetic or Real) We Should Rely for Soft Biometric Classification*, IEEE/CVF Winter Conf. on Applications of Computer Vision (**WACV 2025**), h5-index: 109.

* denotes equal author contribution.

Startup & Grants

1. **Startup: GUARDMYPRIVACY Private Limited** 2026–Present
A privacy-tech startup founded to commercialize research on protecting personal data from unauthorized AI usage and data exploitation.
Founders: Atul Kumar, Dr. Akshay Agarwal
2. **Grant: AIC SEED Plan-E (Cohort-1)** IISER Pune, 2025
Competitive startup grant awarded under the initiative **Convert Your PhD Journey into a Startup** to support the development and commercialization of GUARDMYPRIVACY.
Awardees: Atul Kumar, Dr. Akshay Agarwal
3. **Grant: IICE Student Innovation Grant (SIG 1.0)** IISER Bhopal, 2025–26
Received **three consecutive innovation grants** under the SIG 1.0 program over every 6-month period to translate privacy-preserving AI research into real-world technologies and product development.
Awardees: Atul Kumar, Dr. Akshay Agarwal, S. Aarthi

Scientific Duty

- **Reviewer** for premier international conferences, including **CVPR**, **ECCV**, **NeurIPS**, and **IJCB**.
- **Reviewer** for international workshops, including **BMVC Workshops** and the **Global South in AI Workshop @ ICML**.

IPR & Patents

1. **Method and System for Generating Unlearnable Images Using Non-Linear Multiplicative Perturbations**
Indian Patent No. 522247 *Granted: 08 August 2026*
Inventors: **Atul Kumar**, Dr. Akshay Agarwal
A privacy-preserving framework that generates visually faithful yet AI-resistant images using non-linear multiplicative perturbations to protect identity from unauthorized machine learning models.

Conferences Workshops

- Attended the **India AI Summit 2026**, a national forum bringing together researchers, industry leaders, and policymakers to discuss the future of Artificial Intelligence.
- Attended the **Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP 2025)**, India's premier conference in Computer Vision and Pattern Recognition.
- Attended and volunteered at the **Indian Control Conference (ICC 2024)**, contributing to the organization of plenary talks and technical sessions.

Technical Skills

Research Areas	Privacy-Preserving AI, Trustworthy Machine Learning, Computer Vision, Biometric Security, Machine Unlearning
AI & Machine Learning	Transformers, GANs, Diffusion Models, Vision-Language Models (VLMs), Representation Learning, Adversarial Learning
Programming Frameworks & Libraries	Python, C++, Java, Data Structures & Algorithms PyTorch, TensorFlow, OpenCV, NumPy, Pandas, scikit-learn, Matplotlib
Data Engineering	Apache Spark, Apache Hadoop, SQL, Database Management Systems
Web & Cloud	Django, HTML, CSS, Google Cloud Platform (GCP)

Achievements & Honours

- Conferred the prestigious **41st Madhya Pradesh Young Scientist Award** for outstanding contributions to AI and privacy-preserving research.
- Awarded the prestigious **ACM-IARCS Travel Grant 2026** to present research at the **European Conference on Computer Vision (ECCV 2026)**.
- Secured **4th Position Worldwide** in the **Gen μ 2.0 Face Unlearning Challenge** at the **U&ME Workshop, ECCV 2026**.
- Secured **6th Position Worldwide** in the **ACM Multimedia (ACMMM) AADD Challenge 2026** as part of Team **GhostChamber**.
- Honoured as the **Best Teaching Assistant**, IISER Bhopal (2025).
- Awarded **Best Poster Presentation**, DSE & Engineers' Day, IISER Bhopal (2025).
- Awarded **Best Poster Presentation**, DSE & Engineers' Day, IISER Bhopal (2024).
- Qualified **GATE 2024** in Computer Science (CS) and Data Science & AI (DA).
- Qualified **UGC NET-JRF 2023**.

Outreach & Activities

- Volunteered in organizing the Impact Summit on Democratizing AI, IISER Bhopal, *February 2026*.
- Volunteered in organizing the Workshop on Next-Generation AI on the occasion of DSE Day, IISER Bhopal, *November 2025*.
- Volunteered in organizing Plenary Talks on Cutting-Edge Technologies & Startups on Engineers' Day, IISER Bhopal, *September 2025*.
- Volunteered in organizing the Indian Control Conference (ICC 2024), IISER Bhopal, assisting in the management of plenary talks and technical sessions, *December 2024*.